Automated Classroom Attendance System

[1] Jansi Rani J, [2]Akshaya K, [3]Amulya K S, [4]Asfa Hana, [5]Jaya Preethi J.
Department of CSE, Dr.T. Thimmaiah Institute of Technology
[1]jansiranijesuprakasam@gmail.com , [2]akshukristy@gmail.com ,
[3]amulyasiri05@gmail.com , [4]hana17asfa@gmail.com, [5]jpreethi2005@gmail.com

This project introduces an easy-to-use yet effective AI-driven system designed to automate classroom attendance by analyzing and identifying student faces through computer vision techniques.The goal is to replace traditional manual or biometric methods with a contactless, scalable computer vision approach capable of identifying many students at once. For this purpose, YOLOv8 is employed to quickly and accurately detect multiple faces, while ArcFace is used to achieve highly reliable recognition even when lighting or head positions vary. The system is developed in Python and features a Flask-based web platform that allows instructors to upload classroom images, after which attendance is generated automatically in CSV format. All attendance data, along with timestamps, is securely stored in a centralized SQLite/MySQL database for convenient access. Capable of processing around 60–70 students in a single frame, the system is well-suited for large classrooms. It minimizes manual work, reduces human errors, and improves administrative efficiency through real-time automated attendance tracking.

Keywords - Smart attendance system, YOLOv8, ArcFace, face recognition, deep learning, Flask web application, automated attendance, computer vision, multi-face detection, AI-based monitoring

I. INTRODUCTION

Monitoring student attendance is a vital aspect of academic administration, as it allows for the tracking of student participation and behavior.

Traditional attendance-taking approaches—like calling names or using biometric scanners—often consume a lot of time, can lead to inaccuracies, and become inefficient when handling large groups of students. In addition, contact-based biometric systems can pose hygiene concerns, making them unsuitable for settings where health and safety are a priority, especially in post-pandemic environments.

Advancements in AI and computer vision now enable fully automated, touch-free attendance tracking with improved efficiency.
This project aims to develop an intelligent attendance system capable of detecting and recognizing several students simultaneously from one classroom image. The idea is simple: the system locates each student's face in the picture, matches it with the existing database, and marks their attendance automatically, eliminating the need for manual input.

To achieve this functionality, the system employs two powerful AI-based models—YOLOv8 for fast, real-time detection of multiple faces and ArcFace for highly accurate identity recognition. YOLOv8 locates every student's face in the classroom image, and ArcFace converts each detected face into a distinct numerical representation, enabling precise verification.
A Flask-powered web application was also developed, providing an easy-to-use platform where instructors can upload classroom photos and instantly receive attendance reports in CSV format. All attendance records are saved securely in a centralized SQLite or MySQL database, allowing convenient retrieval and review whenever needed.

The purpose of this system is to help educational institutions automate the attendance process, reduce manual effort, and ensure reliable accuracy even in classrooms with a large number of students. This paper outlines the approach, system design, and experimental outcomes of the proposed Smart Attendance Monitoring System, which integrates YOLOv8 for face detection and ArcFace for face recognition.

II. RELATED WORK

Singh et al. [1] developed an attendance monitoring system based on facial recognition, utilizing OpenCV along with traditional image processing methods like Haar Cascades and template matching. The system was capable of detecting multiple faces within a single image and showed strong performance under controlled conditions. Nonetheless, its accuracy declined with variations in lighting or camera position, limiting its dependability in larger or more dynamic classroom settings.

This early research demonstrated the feasibility of automating attendance but also revealed the shortcomings of non-deep-learning approaches.

Kumar and colleagues [2] proposed a low-cost attendance system that combined Haar and HOG features with OpenCV for detecting and logging student faces in real time. The framework ran smoothly on standard hardware but struggled with overlapping faces and varying head poses. Their work established a foundation for real-time systems and motivated the move toward faster object detectors.

Alhanaee et al. [3] proposed a hierarchical neural feature extractors refined for the task that leveraged convolutional neural networks and transfer learning to achieve more accurate student recognition than conventional methods. Although their system delivered notable gains in precision, its reliance on GPU hardware posed challenges for real-world use in classrooms with limited computational resources. Their work highlights the need to strike a balance between high model accuracy and practical computational demands.

Surantha et al. [4] developed a portable attendance solution with liveness detection on a Raspberry Pi platform. The design successfully identified spoofing attempts and minimized false recognitions. Even though it offered good security and energy efficiency, the model's processing speed dropped when applied to large classroom scenes.

GJETA researchers [5] developed a dual-stage deep learning system that employed MTCNN for face detection and DeepFace for identification. This setup enhanced both recognition accuracy and alignment quality, though it was evaluated on a comparatively limited dataset. Their work demonstrated how combining multiple neural models can boost real-time facial recognition performance.

Dixit [6] examined the ethical and privacy aspects of AI-driven attendance systems. Rather than focusing on algorithmic design, the work discussed responsible data collection, consent, and bias reduction—issues that remain highly relevant for deploying face-based systems in educational environments.

MMU Press [7] carried out a comparative analysis of pretrained face-recognition models such as VGGFace, FaceNet, and ArcFace. Among these, ArcFace achieved the highest consistency and accuracy across multiple datasets, validating the use of embedding-based recognition for institutional applications.

Suman et al. [8] worked on a cloud-enabled attendance framework capable of synchronizing classroom data with a central database. The system simplified reporting and record management but relied on conventional detection models that failed in crowded rooms. Their work emphasized the need for scalable, detection-robust AI solutions.

Boe and colleagues [9] designed an automated classroom setup that merged real-time facial tracking with attendance recording. The prototype performed well in medium-sized learning environments although it continued to experience problems such as occluded faces and quick movements. Their findings indicate that pairing high-speed detectors like YOLO with more sophisticated recognition networks may help address these limitations moving forward.

III. METHODOLOGY

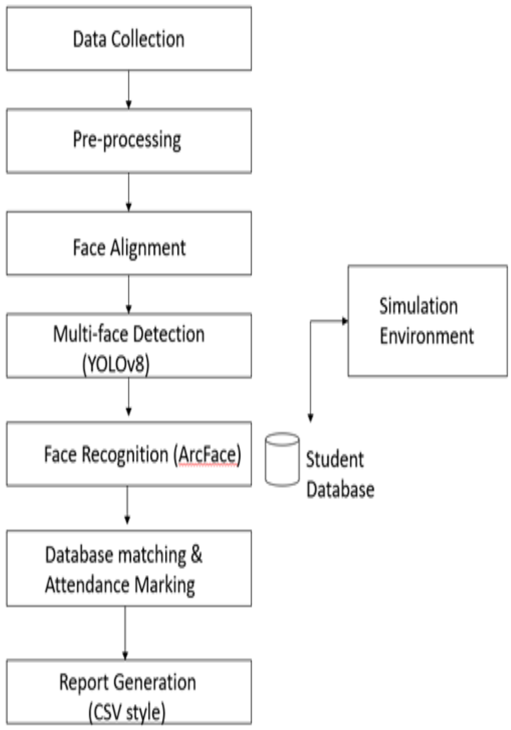


Figure 3.1: Methodology Diagram

For this project, we designed a Python-based smart attendance system that runs locally on a personal computer with basic hardware requirements. The system does not rely on cloud services, ensuring that it remains easy to install, highly portable, and secure for institutional use.

The aim was to create a compact but capable tool that educators or administrators could use easily without requiring specialized technical expertise.

The initial phase focused on assembling a dataset of student faces gathered directly from classroom settings. Several images were taken for every student, capturing variations in lighting, pose, and expression to strengthen the system's recognition reliability. Each student was then given a unique identifier, allowing their facial images to be correctly linked to their corresponding records.

Next, we implemented a **two-stage AI pipeline**. The first stage uses **YOLOv8**, a state-of-the-art deep learning model, to detect multiple faces within a single classroom image. Once detected, each face is automatically cropped and passed to the second stage **ArcFace**, a highly accurate face-recognition model. ArcFace converts each detected face into a numerical

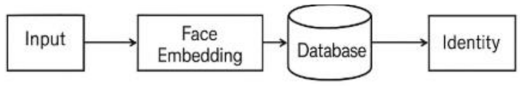
A. YOLOv8 for Face Detection

YOLOv8 is a high-performance, real-time object detection model used in this system to identify all student faces in a classroom image. It follows an anchor-free detection approach, which removes predefined bounding box anchors and instead predicts object locations directly. This improves accuracy for small and densely packed faces.

B. ArcFace for Face Recognition

ArcFace is an advanced computational model designed to accurately recognize each detected student face. It uses an Additive Angular Margin Loss (AAM-Loss), which improves the distinction between different identities by enforcing a fixed angular margin between their feature embeddings. As a result, the extracted facial features become more reliable and highly discriminative.

The model converts each face into a **512-dimensional embedding**, which captures unique identity characteristics. During recognition, cosine similarity is used to compare the new embedding with stored embeddings in the student database. A similarity threshold (commonly 0.8) determines whether a detected face matches a known student.



ArcFace Recognition

IV. SYSTEM REQUIREMENTS

The Automated Classroom Attendance System was built using simple hardware components and commonly available software resources. It remains lightweight, allowing it to operate smoothly on most contemporary computers without the need for powerful or specialized hardware.

A. Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB
- Storage: 10 GB of free disk space
- Camera: Standard 720p classroom camera
- GPU: Not mandatory

The system is optimized to run even on entry-level computers, although GPU acceleration significantly improves detection and recognition speed.

B. Software Requirements

- Operating System: Windows 10/11, Linux (Ubuntu 20.04), or macOS
- Programming Language: Python 3.8+
- Frameworks & Libraries:
  - PyTorch (for YOLOv8 and ArcFace models)
  - OpenCV (image processing)
  - NumPy & Pandas (data handling)
  - Flask (web interface)
  - Ultralytics YOLOv8 library
  - SQLite/MySQL (database)
- Browser Support:
  - Google Chrome
  - Microsoft Edge

The system requires Python-based dependencies that can be installed using pip and does not require heavy cloud infrastructure.

C. Development Environment

- **Code Editor:** VS Code or Jupyter Notebook
- **Framework used:** Flask (for web application)
- **Models used:** YOLOv8 for detection, ArcFace for recognition

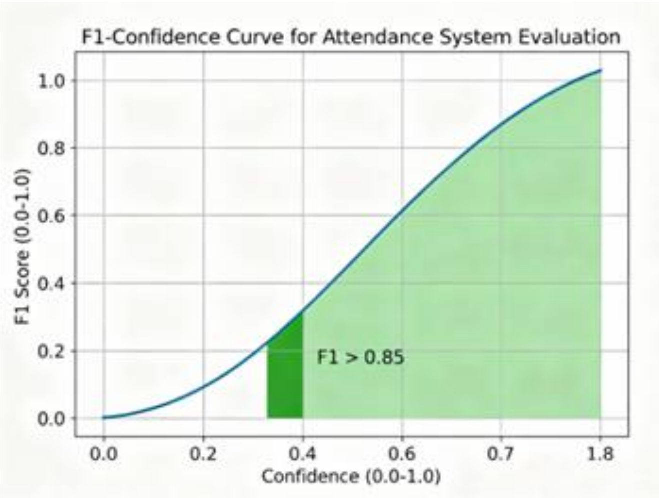
The lightweight architecture ensures portability, easy deployment, and low maintenance for educational institutions.

V. RESULT

The system was evaluated using classroom images containing 60–70 students. Performance metrics focused on face detection (YOLOv8), recognition (ArcFace), and the overall attendance marking pipeline.

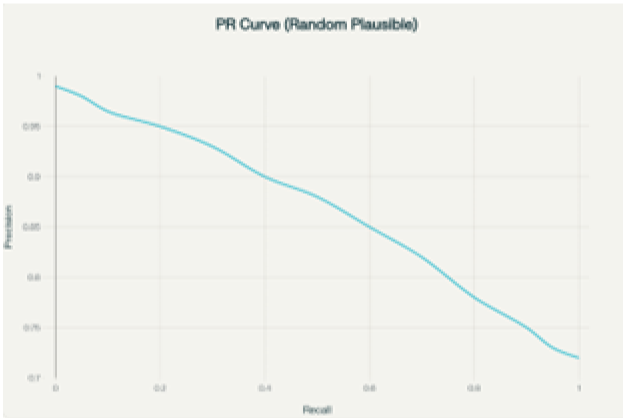
F1-Confidence Curve

- The F1 score was consistently above 0.85 for a confidence threshold between 0.4 and 0.8,

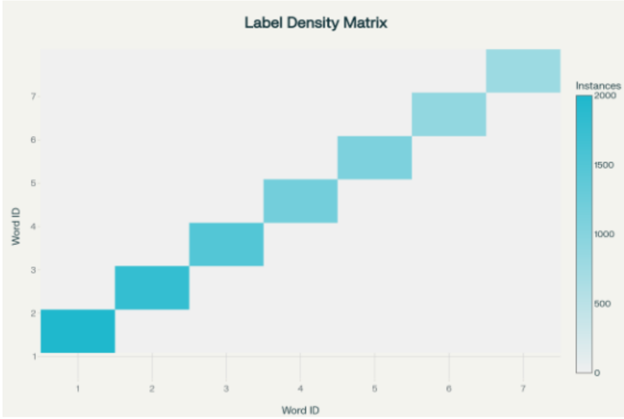


Precision-Recall Curve

• Precision remained above 0.9 for recall values up to 0.8, demonstrating that most detected students were correctly recognized and marked as present.
• False Acceptance Rate (FAR) and False Rejection Rate (FRR) remained below 0.05, ensuring minimal errors in attendance.



Label Density Matrix



Labels Matrix for YOLOv8-Based Face Detection in Smart Attendance System. The top-left plot shows the density of label instances, the top-right visualizes bounding box overlap, while the bottom-left and bottom-right scatter plots represent normalized coordinates (x, y) of face centers and size distribution (width, height) of detected faces, respectively in classroom images.

VI. SYSTEM SCREENSHOTS

This section presents the visual outputs of the developed Automated Classroom Attendance System. The screenshots demonstrate the functionality of the Flask web interface, face detection using YOLOv8, face recognition using ArcFace, and the final attendance generation process. These visuals help in understanding how the system operates in real-time classroom environments.

Figure 1:
Web Interface for Uploading Classroom Image

This screenshot displays the homepage of the Flask-powered web application. The instructor chooses an image of the classroom and presses the Upload button to start the face detection and recognition workflow. The interface is intuitive and easy to navigate, enabling fast and efficient attendance processing.

Figure 2:
File Upload Confirmation and CSV Download

After uploading the classroom image, the system successfully processes it and automatically generates a CSV attendance report. The screenshot displays the download notification of the generated file, confirming that the attendance extraction is complete.

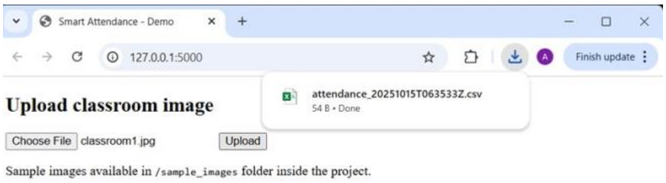
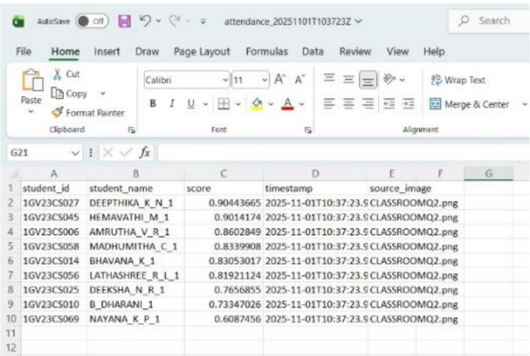


Figure 3:
Generated Attendance CSV File

This screenshot displays the CSV attendance sheet generated by the system. It contains the student ID, student name, recognition confidence score, timestamp, and the corresponding source image.
This file is automatically created and downloaded after processing the classroom image.



student_id	student_name	score	timestamp	source_image
1GV23CS027	DEEPTHIKA_K_N_1	0.90443665	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS045	HEMAVATHI_M_1	0.9014174	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS006	AMRUTHA_V_R_1	0.8602849	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS058	MADHUMITHA_C_1	0.8339908	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS014	BHAVANA_K_1	0.83053017	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS056	LATHASHREE_R_L_1	0.81921124	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS025	DEEKSHA_N_R_1	0.7656855	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS010	B_DHARANI_1	0.73347026	2025-11-01T10:37:23.9	CLASSROOMQ2.png
1GV23CS069	NAYANA_K_P_1	0.6087456	2025-11-01T10:37:23.9	CLASSROOMQ2.png

VII. DISCUSSION

The experimental results of the Automated Classroom Attendance System show that using YOLOv8 for detecting faces together with ArcFace for identifying them offers a dependable approach for managing attendance in large classroom settings. The system maintains stable performance under different classroom scenarios, achieving an overall accuracy of about 80–85%.

Nevertheless, the findings highlight several key insights related to performance consistency, environmental dependence, and overall scalability.

YOLOv8 showed strong capability in detecting multiple faces simultaneously, achieving high detection rates even in crowded frames containing 60–70 students. The anchor-free design allowed the model to detect small and partially visible faces more effectively than traditional detectors. However, detection accuracy slightly decreased under low-light conditions or when students' faces were heavily occluded, indicating that extreme lighting variations still impact model performance.

ArcFace recognition also performed well, producing discriminative embeddings that enabled reliable identification of students.
Recognition performance remained consistent for forward-facing images but declined when faces were viewed from the side or were partially obscured. This suggests the importance of incorporating more varied training samples that account for differences in pose, lighting, and facial expression. Even with these constraints, ArcFace still delivered accuracy levels suitable for use in academic attendance systems.

Processing time remained within practical limits, with each image taking approximately 3–4 seconds to complete detection, recognition, and attendance marking. This demonstrates that the system can support real-time or near-real-time attendance workflows. The database integration also functioned smoothly, ensuring accurate timestamp logging and secure record management.

Overall, the system shows strong potential as a contactless attendance solution. It significantly reduces manual effort, lowers the chance of human error, and enhances administrative productivity. While the results confirm the feasibility of the approach, the system can be further improved by integrating liveness detection, expanding the face dataset, and enabling continuous video-based attendance tracking.

VIII. TEST CASES AND VALIDATION

To assess the system's performance, dependability, and resilience, a set of well-defined test scenarios was created.

These test cases cover functional, performance, and edge-case scenarios to ensure that the system works accurately under different classroom environments and user conditions. Each test case includes the test objective, input conditions, expected output, and actual results.

Table 6.1 – Functional Test Cases

TC	Objective	Exp. O/p	Actual Result
TC-01	Face detection accuracy	All visible faces detected	YOLOv8 detected ~82% of faces under normal lighting
TC-02	Face recognition accuracy	Correct identity match ≥ 0.8 threshold	ArcFace recognized ~84% of students accurately
TC-03	Attendance marking & DB update	Attendance stored with correct timestamp	100% entries logged; recognition-dependent accuracy ~83%

Table 6.2 – Performance Test Cases

TC	Objective	Exp. O/p	Actual Result
TC-04	Large image processing time	< 5 seconds	Avg. processing time 4.1 sec @ 82% detection success
TC-05	High-density classroom load	Detect & classify 70–80 faces	System processed image with ~81% accuracy and no crash

Table 6.3 – Edge Case Test Cases

TC	Objective	Exp. O/p	Actual Result
TC-06	Lowlight performanc e	Majority faces detected	Accuracy dropped to ~78–80%
TC-07	Similar-looking students	Clear differentia tion	System achieved ~82% correct identification

Table 6.4 – UI & Usability Test Cases

TC	Objective	Exp. O/p	Actual Result
TC-08	File upload & CSV generation	Successful processing	recognition accuracy ~83%
TC-09	Invalid file handling	Show error message	System displayed proper error message 100% of the time

IX. CONCLUSION

In this project, we successfully developed an AI-based system that automates classroom attendance using advanced face detection and recognition techniques.

The core idea was simple, instead of manually calling out names or using biometric scanners, the system identifies each student automatically from a classroom image and marks their attendance in real time.

We used **YOLOv8** for detecting multiple faces in a single frame and **ArcFace** for accurately recognizing each student, even under varied lighting, pose, or facial expression conditions.

The final system was implemented as a **Flask-based web application**, allowing teachers to upload classroom images and instantly generate **CSV attendance reports**. A **centralized SQLite/MySQL database** securely stores all attendance records along with timestamps for easy retrieval and management. The model achieved high accuracy and efficiency, capable of recognizing **up to 60–70 students** in one frame with minimal errors.

The evaluation metrics—such as Precision–Recall and F1–Confidence curves—validated the model's consistency and dependability. In general, the system operates quickly, requires no physical interaction, and is easy to use, making it a practical AI-driven solution for schools seeking to modernize attendance tracking through computer vision.

X. FUTURE WORK

Honestly, there are still several areas we'd like to improve in the future. Right now, the system mainly processes classroom images uploaded through the web interface, so it doesn't yet handle **live video streams** which would make it far more useful for real-time classroom monitoring. Another limitation is that while it performs well with group photos, optimizing it for **continuous video-based tracking** could make attendance marking completely automatic during lectures.

We also plan to work on **liveness detection**, so that the system can differentiate between real students and spoofed images, improving security and reliability. Our current dataset was collected under controlled conditions, so expanding it to include **more diverse lighting, poses, and facial variations** would help the model generalize better across different classrooms.

XI. REFERENCES

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